

36 Reactions on surfaces and Review of catalysis

Why Are We Learning This?

In previous lectures, we examined chemical reactions at the molecular level and developed reaction mechanisms for gas-phase systems. However, many of the world's most important chemical processes do not occur solely in the gas or liquid phase. Instead, they occur on catalyst surfaces.

Surface reactions are responsible for the production of fuels, polymers, fertilizers, pharmaceuticals, and countless other products. By providing an alternative reaction pathway, catalysts make reactions proceed faster and often make otherwise impractical reactions economically feasible.

This lecture introduces the fundamental molecular processes involved in heterogeneous catalysis and shows how these ideas apply to one of the most important catalytic processes ever developed: ammonia synthesis via the Haber-Bosch process.

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



Reaction Progress
(Conversion)



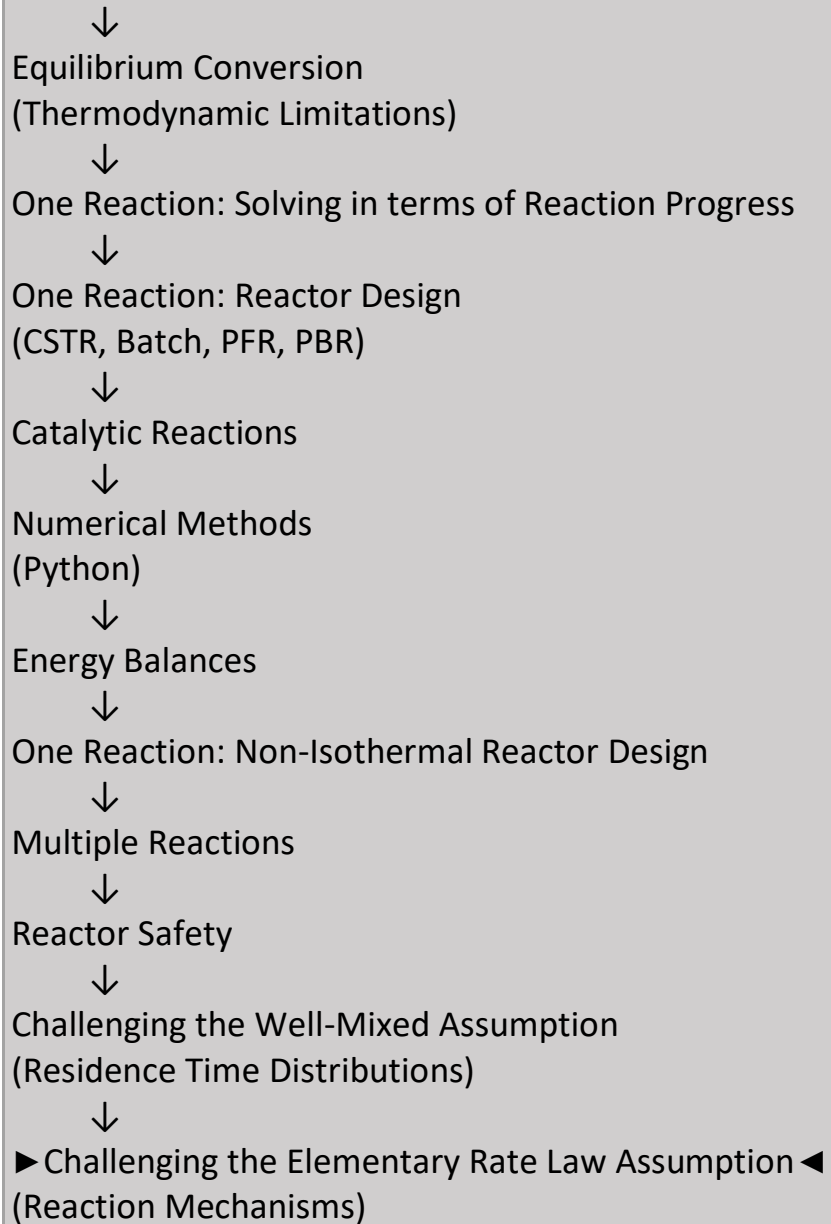
Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)



Rate Laws
(Kinetics)



Equilibrium
(Reversible Reactions)



Learning Objectives

By the end of this lecture, you should be able to:

- Explain how catalysts affect reaction rates.
- Describe the role of activation energy in chemical reactions.
- List the major steps in a catalytic reaction mechanism.
- Define adsorption and desorption.

- Explain why catalyst surfaces are chemically active.
- Describe the role of iron catalysts in ammonia synthesis.
- Explain why catalytic ammonia synthesis is more favorable than gas-phase ammonia synthesis.
- Interpret surface-reaction mechanisms written using adsorbed species.

Big Idea

Catalyst Surface



Provides Alternative Reaction Pathway



Lower Activation Energy



Faster Reaction

Catalysts do not change equilibrium. They change how reactions reach equilibrium.

This lecture is based on the [Nobel Lecture by Gerhard Ertl](#)

A chemical reaction involves

This involves

What Must Happen for a Reaction to Occur?

At the molecular level, a chemical reaction requires:

Old Bonds Break

↓

Atoms Rearrange

↓

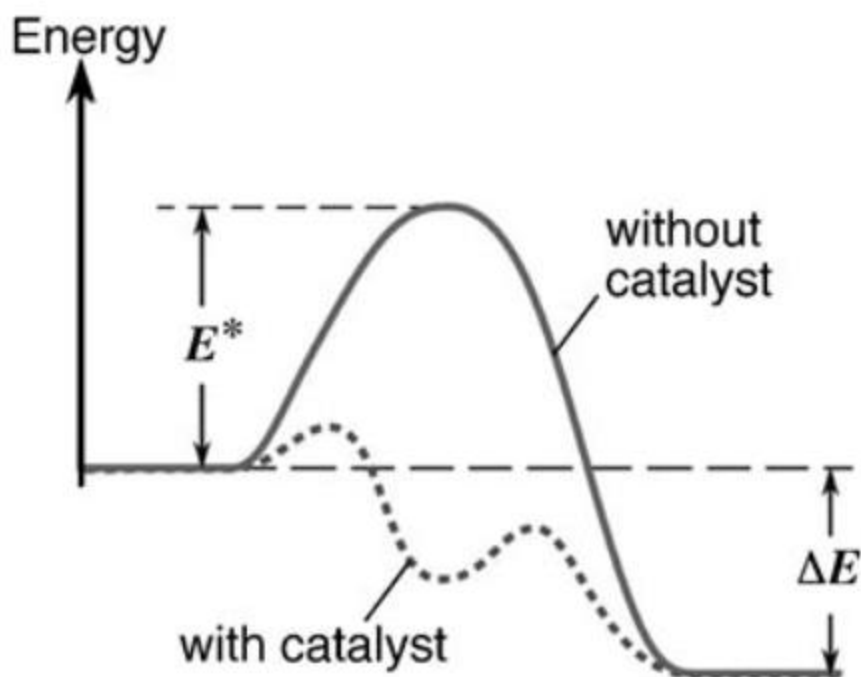
New Bonds Form

Every chemical reaction involves this fundamental molecular process.

For the reaction to occur, enough energy must be available (or supplied) to overcome

The rate is proportional to $e^{\frac{-\Delta E^{act}}{RT}}$; hence, the rate can be increased by

A catalyst decreases the activation energy by



What Is Activation Energy?

Before reactants can be converted into products, they must overcome an energy barrier known as the **activation energy**.

Physical Interpretation

Reactants

↓

Activation Barrier

↓

Products

Only molecules with sufficient energy can cross this barrier and react. Molecules that do not have enough energy simply collide and separate without reacting.

Why Temperature Matters: Increasing the temperature increases the fraction of molecules that possess enough energy to overcome the activation barrier.

Higher Temperature



More Energetic Molecules



More Successful Collisions



Faster Reaction Rate

This is why most reaction rates increase rapidly with temperature.

Key Insight

Activation Energy



Controls How Many Molecules Can React

A reaction may be thermodynamically favorable, but it will proceed slowly if its activation energy is large.

In catalysis, the reaction happens on the

This happens in several steps:

1. Reactant molecules "adsorb" to the catalyst surface. Adsorption is

1. Reactant molecules fully or partially decompose on the catalyst surface. For example:

1. Atoms and/or molecules on the catalyst surface react with each other. For example:

1. Product molecules "desorb" from the catalyst surface. Desorption is

The catalyst surface readily binds with atoms and molecules because

What Does a Catalyst Actually Do?

A catalyst does **not**: Create Energy, Change Equilibrium, Change the Overall Thermodynamics

Instead, a catalyst provides an alternative reaction pathway with a lower activation energy.

Physical Interpretation

Uncatalyzed Reaction



Large Activation Barrier

Catalyzed Reaction



Smaller Activation Barrier

Because the activation barrier is lower, a much larger fraction of molecules can react successfully at the same temperature.

Why Catalysts Increase Reaction Rates

Lower Activation Energy



More Effective Collisions



More Molecules React



Faster Reaction Rate

Key Insight

Catalyst



Changes the Path



Lowers Activation Energy



Increases Reaction Rate

A catalyst makes a reaction easier to perform, but it does not change the reactants, products, or equilibrium position. It simply provides a faster route between them.

The Four Fundamental Steps of Heterogeneous Catalysis

Most heterogeneous catalytic reactions occur through a sequence of surface processes rather than a single reaction step.

Step 1: Adsorption

Reactant molecules attach to the catalyst surface.

Reactants in Fluid Phase



Adsorbed Reactants

Step 2: Activation

Adsorption often weakens existing chemical bonds, making the reactants more chemically reactive.

Adsorbed Species



Bond Weakening



Intermediate Species

Step 3: Surface Reaction

Adsorbed atoms and molecules react with one another on the catalyst surface to form products or intermediates.

Adsorbed Reactants



Surface Reaction



Adsorbed Products

Step 4: Desorption

The product molecules leave the catalyst surface and return to the gas or liquid phase.

Adsorbed Products



Desorption



Products in Fluid Phase

Overall Picture

Adsorption



Activation



Surface Reaction



Desorption

These four steps form the foundation of most heterogeneous catalytic reaction mechanisms. Although the details vary from system to system, most catalytic processes can be understood in terms of this general sequence.

Why This Matters

A catalyst does not typically make molecules react immediately. Instead, it provides a surface where reactants can adsorb, activate, encounter other reaction intermediates and hence react more easily, and then leave as products. This alternative pathway often has a much lower activation energy than the corresponding gas-phase reaction.

An important application of catalysis is the ammonia synthesis reaction, $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$. In the first part of the 20th century, the world was facing starvation due to an increasing population and depletion of fertilizers. The Haber-Bosch process was designed in the 1930s to combat this by generate ammonia from air.

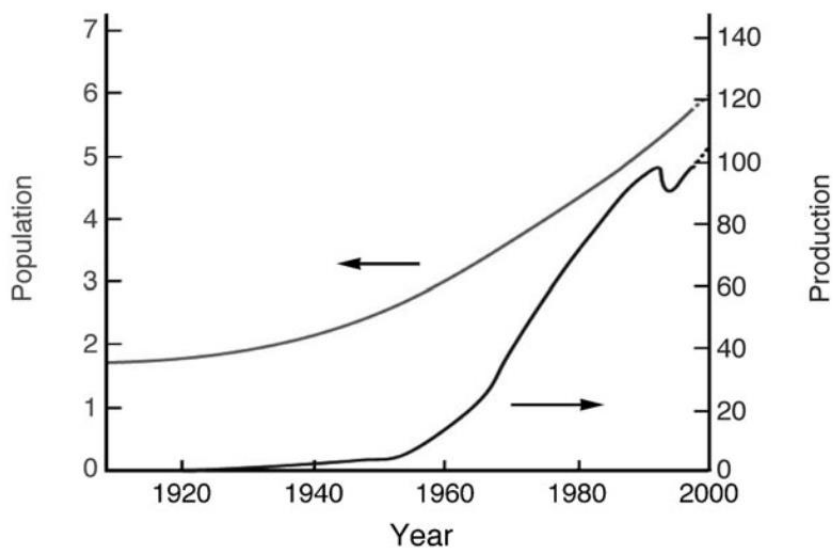


Figure 3. Variation of the world population (in 10^9) and the ammonia production (in 10^6 metric tons of nitrogen) during the 20th century.^[7]

Why Is Ammonia Synthesis So Important?

Ammonia synthesis is one of the most important catalytic processes ever developed. Ammonia is used to produce: Fertilizers, Chemicals, Industrial Products

The Haber-Bosch process dramatically increased agricultural productivity and helped support modern population growth.

Historical Significance

Atmospheric Nitrogen

↓

Catalytic Ammonia Production

↓

Nitrogen Fertilizers

↓

Increased Food Production

The Haber-Bosch process utilizes

Industrial ammonia synthesis catalysts comprise nanometer sized particles with surface areas of ~

Iron oxide present under reaction conditions is reduced to metallic iron by incorporating

The particles are stabilized against agglomeration by attaching to

The main utility of the Fe catalyst is to

Why Is Iron Used?

One of the greatest challenges in ammonia synthesis is breaking the extremely strong nitrogen triple bond: $\text{N}\equiv\text{N}$

This bond is among the strongest chemical bonds commonly encountered in chemical reactions, making nitrogen relatively unreactive in the gas phase. The primary role of the iron catalyst is to adsorb nitrogen molecules and weaken the $\text{N}\equiv\text{N}$ bond, making it easier for hydrogen atoms to react with nitrogen and form ammonia.

Physical Interpretation

N_2 Molecule



Adsorbs on Iron Surface



$\text{N}\equiv\text{N}$ Bond Weakens



N_2 is activated, forming N



N is stabilized by the surface



Hydrogen Adds to Nitrogen



NH_3 Formation Becomes Easier

Why This Matters

Without a catalyst: N_2 Must Break Apart in the Gas Phase which requires a large activation energy.

With an iron catalyst:

N_2 Adsorbs on the Surface



Bond Is Activated



Reaction Proceeds Through Lower-Energy Steps

As a result, ammonia can be produced at industrially useful rates.

Key Insight

Iron Does Not Supply Energy



Iron Provides an Easier Path



Nitrogen Activation Becomes Possible



Ammonia Synthesis Accelerates

The most important function of the iron catalyst is not to create ammonia directly, but to make nitrogen activation and bond breaking energetically feasible.

On the catalyst surface, this looks like:

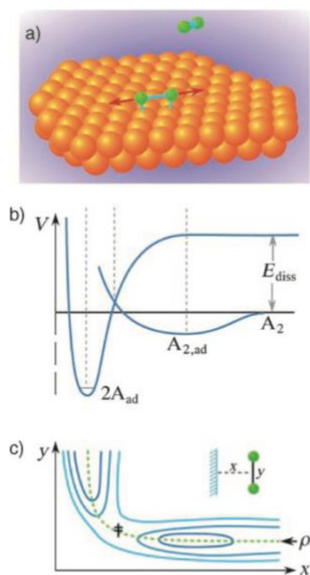
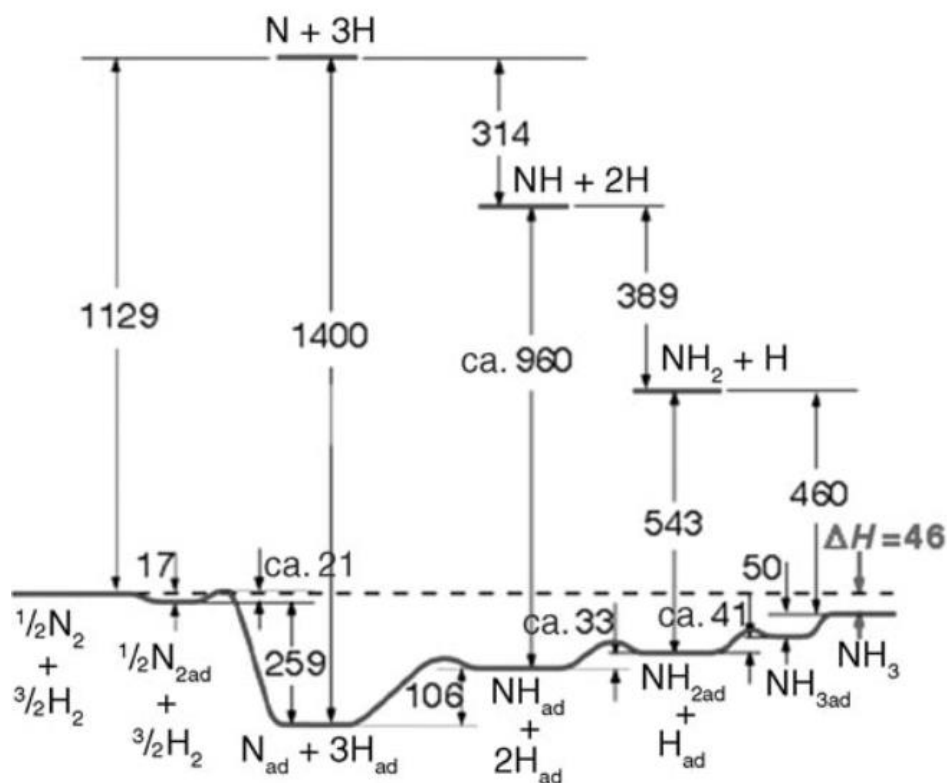


Figure 5. a) Illustration of the process of dissociative chemisorption of a diatomic molecule; b) one-dimensional Lennard-Jones diagram; c) two-dimensional representation of equipotential lines as function of the distance x of the molecule from the surface and the separation y between the two atoms.

It is much more energetically feasible than in the gas phase:



Why Is the Surface Route Easier?

In the gas phase: N_2 Must Dissociate Without Assistance which requires a large amount of energy.

On the catalyst surface:

Adsorption

↓

Bond Weakening

↓

Stepwise Reactions

make the process much more energetically favorable.

Key Insight

Gas Phase



Large Activation Barrier

Catalyst Surface



Lower Activation Barrier

The catalyst provides a molecular pathway that is inaccessible in the uncatalyzed system.

From this diagram, we can write the mechanism for the Fe-catalyzed reaction:

In this notation, "ad" means

What Does "ad" Mean?

The notation: N_{ad} , H_{ad} , and NH_{ad} indicate an adsorbed species.

Interpretation

N_{ad} means: Nitrogen Bound to the Catalyst Surface

Similarly: H_{ad} means: Hydrogen Bound to the Catalyst Surface

These species are not in the gas phase—they reside on the catalyst surface and participate in surface reactions.

Summary and Key Takeaways

Today we reviewed the fundamentals of heterogeneous catalysis and surface reactions.

Key Ideas

- ✓ Chemical reactions require bond breaking and bond formation.
- ✓ Activation energy controls reaction rates.
- ✓ Catalysts increase reaction rates by lowering activation energy.
- ✓ Heterogeneous catalytic reactions occur on surfaces.
- ✓ Most catalytic mechanisms involve adsorption, activation, surface reaction, and desorption steps.
- ✓ Iron catalysts facilitate ammonia synthesis by activating nitrogen molecules.
- ✓ Surface reaction mechanisms provide an atomistic description of catalytic chemistry.

Most Important Idea

Catalyst Surface

↓

Adsorb Reactants



Activate Bonds



Promote Surface Reactions



Desorb Products

The central lesson of this lecture is that catalysts accelerate reactions by providing alternative molecular pathways that occur on reactive surfaces, allowing difficult bond-breaking and bond-forming processes to occur much more easily.